

TinyML Badminton Coach

Real-time badminton smash quality classification using IMU data and on-device TinyML inference

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Problem & Motivation

- Badminton smash quality is difficult to judge without coach feedback
- Players may not know whether a swing is strong, weak, or not a valid smash during practice
- Our system uses IMU motion data from the Arduino Nano 33 BLE Sense Rev2 to classify swings into good_smash, weak_smash, or non_smash
- TinyML makes the system portable and allows motion classification directly on the device
- On-device inference avoids cloud processing and can provide fast feedback during training

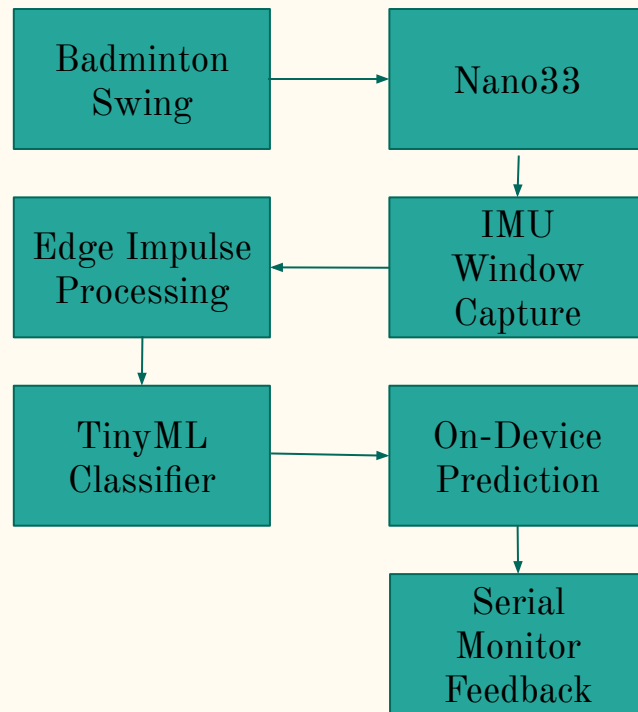
Task, Data, and Metrics

- Classify 1-second badminton motion windows using IMU data from the MCU
- Dataset: 60 total samples
 - 47 training / 13 testing
- Each sample contains accelerometer and gyroscope readings: aX, aY, aZ, gX, gY, gZ
- Target classes: good_smash, weak_smash, and non_smash
- Evaluation metrics: accuracy, confusion matrix, F1 score, model size, RAM / Flash usage, and inference latency

Item	Description
Window	1000 ms
Rate	100 Hz
Inputs	Accel + Gyro
Classes	3
Dataset	60 samples
Split	47 train / 13 test
Platform	Nano 33 BLE Sense Rev2

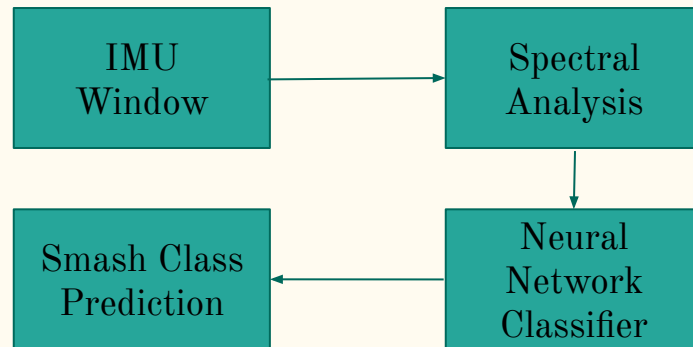
System Overview

- The Nano collects swing motion directly from the onboard IMU
- The trained model runs locally on the board after Edge Impulse deployment
- Output feedback is shown through the Serial Monitor during testing



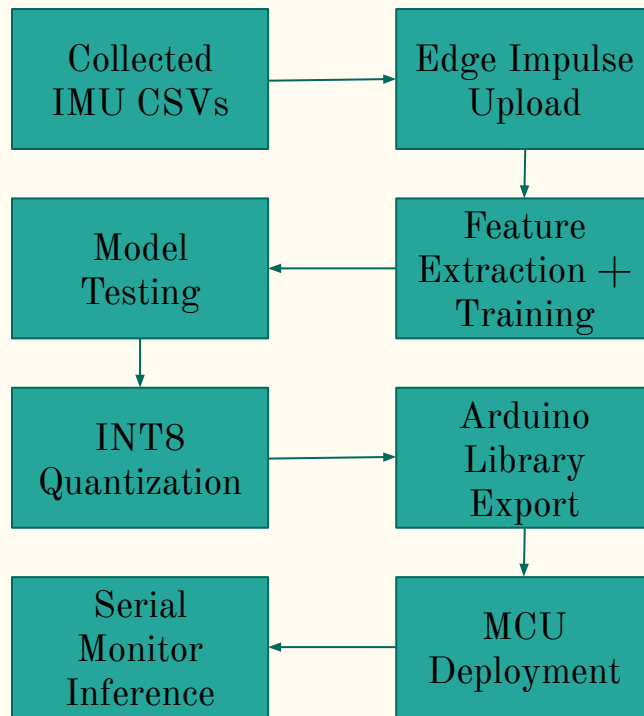
Baseline ML Models

- Baseline pipeline was built in Edge Impulse using IMU time-series data
- The main baseline uses spectral feature extraction before classification
- Training Setup:
 - Window size: 1000 ms
 - Sampling rate: 100 Hz
 - Training cycles: 100
 - Learning rate: 0.005
- Spectral classifier baseline achieved 90.0% validation accuracy, 0.90 weighted F1, 1.4 KB RAM, 15.8 KB Flash, and 1 ms classification latency



TinyML Pipeline

- Edge Impulse was used to train and test the motion classifier
- The selected model was converted into a TinyML-ready Arduino library
- INT8 quantization reduces model size and memory usage for on-device inference
- The final deployed sketch runs inference locally on the Nano 33 BLE Sense Rev2



Advanced Component Integration

1. Spectral features vs raw data pipeline comparison

Pipeline A: Spectral Analysis -> Classification

Pipeline B: Raw Data -> Classification

- Compares frequency-domain features against raw time-series inputs
- Helps determine which pipeline gives better accuracy, latency, and deployment efficiency

2. Float32 vs INT8 quantization comparison

Float32 Model -> INT8 Quantized Model

- Reduces model size and memory usage for TinyML deployment
- Compares accuracy, RAM/Flash usage, and latency before and after quantization

Model	Accuracy	RAM	Flash	Latency
Float32	92.31%	1.6 KB	17.7 KB	35 ms
INT8	92.31%	1.4 KB	15.8 KB	35 ms

Tiny model results

Model / Pipeline	Test Accuracy	F1 Score	RAM	Flash / Model Size	Latency	Deployed?
Spectral Analysis + Classifier	90.0%	0.90	1.4 KB	15.8 KB	1 ms	Yes
Raw Data + Classifier	80.0%	0.80	2.0 KB	26.0 KB	1 ms	No
INT8 Quantized Model	92.31%	0.92	3.2 KB	15.8 KB	35 ms total	Yes

Final model: Spectral Analysis + INT8 Quantized Classifier

Reason: Best accuracy/F1 while using lower memory than the raw-data pipeline

On-device deployment evidence

good_smash:

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.99609
non_smash: 0.00000
weak_smash: 0.00000
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.99609
non_smash: 0.00000
weak_smash: 0.00000
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.99609
non_smash: 0.00000
weak_smash: 0.00000
```

weak_smash:

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.00781
non_smash: 0.04688
weak_smash: 0.94531
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.03516
non_smash: 0.01172
weak_smash: 0.95312
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.04688
non_smash: 0.00391
weak_smash: 0.94922
```

non_smash:

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.02344
non_smash: 0.91016
weak_smash: 0.06641
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.00000
non_smash: 0.69531
weak_smash: 0.30469
```

```
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 87 ms., Classification: 1 ms., Anomaly: 0 ms.):
good_smash: 0.27734
non_smash: 0.71094
weak_smash: 0.01172
```

Key challenges and solutions

Challenge	Solution
Motion threshold tuning	Added live <code>accelMag</code> debug printing to serial monitor to observe real values during swings, then set threshold just below swing peak
IMU never initializing	Missing <code>IMU.begin()</code> call in setup, added error check that stops with a message if init fails
Small dataset (60 samples)	Focused on clean, consistent swings per class rather than volume; Edge Impulse spectral features helped generalize
Class overlap between weak and good smash	Spectral analysis pipeline separated classes better than raw data pipeline (90% vs 80% accuracy)

Conclusion and takeaways

- Built an on-device TinyML classifier for badminton smash motion using Nano 33 BLE Sense Rev2 IMU data
- Spectral features outperformed raw time-series inputs and were selected for final deployment
- Final INT8 model achieved 92.31% test accuracy and 0.92 weighted F1 score

- Future work: collect data from experienced players and add vision-based form analysis

Reference slide

- Edge Impulse Studio – model training, spectral analysis, quantization, and Arduino deployment
- Arduino Nano 33 BLE Sense Rev2 documentation
- Arduino BMI270/BMM150 IMU library documentation
- TensorFlow Lite for Microcontrollers / Edge Impulse EON Compiler
- Team-collected badminton IMU dataset: good_smash, weak_smash, non_smash
- Team-generated Edge Impulse results, deployment screenshots, and Arduino Serial Monitor outputs